# Sarthak kumar seth

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## EDUCATION

**National Institute of Technology Delhi — New Delhi, Delhi**
*Bachelor of Technology in Electrical Engineering* *Graduating 2027*

**The Srijan School — New Delhi, Delhi**
*Senior Secondary* *Completed June 2023*

## WORK EXPERIENCE

**Summer Intern** – DEAL DRDO, Dehradun *June 2025 - July 2025*

- Developed a **Machine Learning–based State of Charge (SoC) estimation** system for Li-ion batteries, integrating data-driven models with **Extended Kalman Filter (EKF)** baselines.
- Built, trained, and optimized ML models using **scikit-learn** and deep learning frameworks, including **Linear Regression**, **Decision Trees**, **Random Forest**, **Gradient Boosting**, **Neural Networks**, and **LSTM**.
- Evaluated model performance using standard regression metrics such as **RMSE**, **MAE**, and $\mathbf{R^2}$, and visualized estimation errors and trends using **Matplotlib** and **Seaborn**.

## PROJECTS

**Smart Expense Tracker** | Smart Expense Tracker |

- Engineered a full-stack finance tracking platform with secure Google OAuth 2.0 authentication, enabling multi-user income and expense management with structured category classification.
- Developed a rule-base budget monitoring engine with automated SMTP email alerts on threshold breach, reducing manual tracking effort by $\tilde{6}0$ and improving proactive budget control.
- Built scalable RESTful APIs and an interactive analytics dashboard generating monthly financial reports, category-wise expenditure insights, and optimized PostgreSQL queries using indexed schema design.
- **Technologies:** Python, Django, Django REST Framework, PostgreSQL, Google OAuth, SMTP, Git

**Battery State of Charge Estimation System** |

- Processed 50k+ time-series samples to estimate battery charge under dynamic operating conditions.
- Trained and compared 5+ models including ensemble methods and LSTM, achieving up to 18% RMSE reduction.
- Validated performance using RMSE, MAE, and $R^2$ across multiple charge–discharge cycles.
- **Technologies:** Machine Learning, Python, scikit-learn, LSTM,jupyter Notebook

**AI-Powered Resume Screening & ATS Ranking System** |

- Built a semantic candidate-ranking engine using Sentence-BERT (all-MiniLM-L6-v2) embeddings and FAISS vector indexing to analyze and rank 100+ resumes against job descriptions based on contextual relevance.
- Developed an ATS scoring pipeline with Top-K candidate retrieval and skill-gap detection, enabling automated shortlisting and reducing manual resume review time by an estimated 70%.
- Integrated Gemini API to generate candidate-fit explanations and resume improvement recommendations, providing actionable feedback on missing skills and keyword coverage.
- **Technologies:** Python, FastAPI, Sentence-BERT (all-MiniLM-L6-v2), FAISS, Gemini API, NLP, Vector Search

## TECHNICAL SKILLS

**Programming Languages:** Python, C++, JavaScript, SQL, MATLAB
**Backend & Web Frameworks:** Django, Flask, FastAPI
**Frontend Technologies:** HTML5, CSS3, Tailwind CSS, JavaScript, Next.js
**Machine Learning:** scikit-learn, TensorFlow, LSTM, XGBoost, Decision Trees, Neural Networks, NLP, Embeddings, Vector Search, Prompt Engineering
**Computer Science Fundamentals:** DSA, Object-Oriented Programming (OOP), OS
**Developer Tools:** Git, GitHub, VS Code, Jupyter Notebook, FAISS, Excel

## EXTRA-CURRICULAR

**Training & Placement Cell, NIT Delhi — Database Head** Dec 2025 – Present
**Unnat Bharat Abhiyan (UBA) Cell, NIT Delhi — General Secretary** Aug 2024 – Present